

LatticeXP2 Brevia is a development board with a small problem—the programmer needs a parallel port, and the USB programming cable is sold separately. The FPGA is a pretty low-end part as compared to the XC3S500E.



Fig. 9: Lattice XP2 Brevia

2. LatticeECP3 Versa

- LatticeECP3 FPGA: LFE3-35EA-8FN484C
- 64Mbit SPI Flash memory
- 1Gbit DDR3
- PCI Express x1 interface
- Four SMA connectors for electrical testing of one full-duplex SERDES channel
- Two RJ45 interfaces to 10/100/1000 Ethernet to GMII
- Expansion connectors for prototyping
- 14-segment alphanumeric display
- Switches, LEDs and displays for demo purposes
- Push-buttons for GPIO and reset, on-board reference clock sources
- Programmed using a mini-USB cable via PCROHS-compliant

LatticeECP3 Versa is one of the cheapest PCI-Express development boards so far. It also has two gigabit Ethernet ports and high-speed serial connectors. It appears that the FPGA device requires a licensed version of the design software.



Fig. 10: Lattice ECP3 Versa

3. Bugblat pif

- A complete FPGA development target—FPGA programming hardware is not needed
- Plenty of on-chip 4-input LUTs—the pif-1200 has 1280, the pif-7000 has 6864
- Plenty of on-chip 9kbit SRAM blocks—the pif-1200 has 7, the pif-7000 has 26

- The FPGA is nonvolatile, with on-chip Flash memory for storing the configuration bit stream
- Up to 256kbits user Flash memory
- Hard-coded I²C, SPI, PLL and timer/counter blocks
- Powered from the Raspberry Pi expansion connector (P1)
- 47 pins of expansion connectors
- Red and green LEDs
- Support software supplied (in Python) for injecting a new configuration into the FPGA
- Example projects supplied, including a project that controls logic inside the FPGA from a web browser

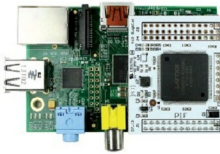


Fig. 11: Bugblat pif board

Bugblat pif is a Raspberry Pi add-on board that provides a MachXO2-1200 or -7000, 17 external I/Os (in addition to those used to communicate with the Raspberry Pi), two LEDs and programming circuitry.

The other popular development boards for Lattice FPGAs are iCEblink40-HX1K Evaluation Kit, TRIFDEV and Bugblat tif.

iCEblink40-HX1K Evaluation Kit comes with a USB programmer, four LEDs, four capacitive touch buttons, configuration PROM, 68 digital I/Os and supposedly some PMOD and Arduino shield compatibility.

TRIFDEV is an Arduino shield that provides 58 extra I/Os, and can also be used as a standalone development board. It has a MachXO2-1200, USB programmer, two buttons and five LEDs.

Bugblat tif is a coin-sized, breadboardable MachXO2-1200 or -4000 board. It has ten I/Os, USB programming (with cross-platform open source software that does not require drivers) and two LEDs.

Development boards for Actel/Microsemi chips

1. Actel IGLOO nano starter kit

- Low-cost AGLN250-VQG100 device
- USB-to-UART connection for HyperTerminal on a PC
- A 20MHz system clock is provided using a low-power, precision oscillator
- Eight LEDs and four switches provide simple inputs and outputs to the system
- All I/Os available for quick connection to external circuitry
- Ability to switch VCORE from 1.2V - 1.5V
- Full current measurement capability of independent I/O banks and



Fig. 12: IGLOO nano starter kit

V_{CC} for accurate power analysis

- Optional board power connector for an external supply, but none is needed when connected to a USB cable
 - All components used on the board are low-power components, including LEDs (2 mA), resets (μ A range) and oscillator (2-3 mA).
- The other popular boards for Actel/Microsemi FPGA chips are Microsemi SmartFusion Evaluation Kit and TrioFlex ST32PA3-AP8.

Microsemi SmartFusion Evaluation Kit features an FPGA advertised as having 200k gates and an integrated 100MHz ARM Cortex-M3, 10/100 Ethernet PHY and on-chip MAC, USB-serial, on-board USB programming interface, OLED display, eight LEDs, two user switches and an indeterminate number of analogue and digital outputs.

TrioFlex ST32PA3-AP8 gives you an Actel A3P060 FPGA and 50MHz oscillator, with an AVR-compatible soft core preloaded in a stamp-sized module.